

Lab: Systems Everywhere -- Mapping Your World

Objective

Learning Goal: Identify and map real-world systems in your daily life, find their boundaries, and discover feedback loops that keep them running.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- Large poster paper or whiteboard
 - Colored markers (at least 4 colors)
 - Sticky notes
 - Your phone or a tablet (optional, for research)
 - Lab journal or notebook
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Part 1: System Spotting (10 min)

Think about the systems you interact with every single day. A **system** is anything with parts that work together inside a boundary. Your school is a system. Your phone is a system. Even your friend group is a system.

Brainstorm with your group: List at least 8 systems you interacted with today. Try to include different types -- social, technological, biological, and organizational.

#	System	Type (Social / Tech / Bio / Org)	What is its boundary?
1			
2			
3			
4			
5			
6			
7			
8			

Write your response here

Part 2: Deep Dive -- Pick One System (15 min)

Choose the most interesting system from your list. Now map it in detail on your poster paper.

Step 1: Draw the Boundary

Draw a big circle or box. Everything INSIDE is part of the system. Everything OUTSIDE is the environment.

Step 2: Identify the Parts

Write each part of the system on a sticky note and place it inside the boundary. For example, if your system is "my school," parts might include: students, teachers, cafeteria, schedule, rules, hallways, etc.

Step 3: Draw the Connections

Use arrows to show how parts affect each other. Use different colors for different types of connections: - **Red** = information flow (messages, signals, data) - **Blue** = resource flow (energy, money, food, materials) - **Green** = feedback loops (when an output becomes an input) - **Black** = rules or constraints

Step 4: Find the Feedback Loops

A feedback loop is when the output of something circles back to affect the input. Example: When the cafeteria runs out of pizza (output), students complain (feedback), and the cafeteria orders more next time (adjusted input).

Write your response here

Part 3: System Surprise (10 min)

Now imagine something unexpected happens to your system. Pick one of these scenarios (or create your own):

- A key part of the system suddenly disappears
- The system gets twice as many inputs as usual
- One feedback loop gets broken

Predict: What would happen? Would the system adapt or break down?

Write your response here

Part 4: Reflection Table

Question	Your Answer
What system did you map?	
How many parts did you identify?	
How many feedback loops did you find?	
What was the most surprising connection?	
How is this system like your brain?	
What would an Active Inference agent need to know about this system?	

Write your response here

Extension Challenge

Compare your group's system map with another group's. Can you find connections BETWEEN the two systems? For example, if one group mapped "the school schedule" and another mapped "the school cafeteria," how do those two systems interact?

This is what Active Inference is all about -- understanding how systems are nested inside bigger systems, and how agents (like you!) navigate them by making predictions.